MILITARY SPECIFICATION

TRANSDUCER, MOTIONAL PICKUP TRU-164/A

This amendment forms a part of MIL-T-83460(USAF), dated 4 February 1975, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1 Delete "first sentence" and add new " Government documents."

* Add: "2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

* Under SPECIFICATIONS, Federal, delete "QQ-P-416" and add: "PPP-B-601 Boxes, Wood, Cleated Plywood."

Under SPECIFICATIONS, Military, delete "MIL-R-83165" and add: "MIL-M-7793/5 METER, TIME Totalizing, Minature, Digital, 28 Volts DC" and "MIL-C-83488 Coating, Aluminum Ion Vapor Deposited."

Add boxed paragraph to bottom center of page:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PAGE 2

* Under STANDARDS, Military, delete "MIL-STD-100," "MIL-STD-810", "MS17321" and add: "DOD-STD-100 Engineering Drawing practices," "MIL-STD-454 Standard General Requirements For Electronic Equipment" and " MIL-STD-810C Environmental Test Methods".

Delete "2.2" and add new

"2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation.

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Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commerical

(Application for copies should be addressed to ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)"

* Add: "2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

PAGE 4

* Add: "3.4.7. Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item."

* Delete "3.5.2" and add: " 3.5.2 Interchangeability. The transducer shall consist of a single unit. All transducers of the same type shall be electrically and mechanically interchangeable. Potentiometric transducers shall not be used. Transducer designs shall be approved by the contracting activity.

PAGE 5

* Figure 1 on the drawing labeled Front delete "MS17321-9" and add: "MIL-M-7793/5-001."

PAGE 7

Delete "Table I reference to NASA-TN-D-822".

PAGE 9

3.5.17 Line 2 delete "MIL-M-7793, and "MS17321-9" and add: "MIL-M-7793/5."

PAGE 10

3.6. STEP h: Delete and add: "h. Random Vibration up to $0.6G_2/Hz$ as specified for jet aircraft in MIL-STD-810C."

PAGE 11

3.8 Line 3 delete "MIL-STD-100" and add: "DOD-STD-100."

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3.11.2 Delete entire paragraph and add the following paragraph

"3.11.2 Steel Parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

* 3.12 Line 2 after manner add: " in accordance with MIL-STD-454, required 9."

* Add: "4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

PAGE 14

Delete "4.4.2".

4.6.13: Delete entire paragraph and add the following:

PAGE 18

"4.6.13 Vibration error test. The transducer shall be subjected to a functional vibration test at room temperature in accordance with MIL-STD-810C, method 514.2, equipment category 6.2, table 514.2 - IIA, procedure IA, figure 514.1 - 2A, curve A. The maximum acceleration power spectral density, Wo, shall be 0.2 G_2/Hz. The transducer shall be tested during vibration for scale error at the test points specified in table I. During vibration, the tolerance shall be 50 percent greater than that specified for room temperature scale error. After this test, the tolerance shall be as specified for room temperature scale error."

4.6.16 Delete entire paragraph and add the following:

4.6.16 Vibration failure test. The transducer shall be subjected to a vibration endurance test at room temperature in accordance with MIL-STD-810C, method 514.2, equipment category 6.2, table 514.2 - IIA, procedure IA, figure 514.2 - 2A, curve A. The maximum acceleration power spectral density, Wo, shall be 0.6 G_2/Hz. The test time per axis shall be two hours. The transducer shall then be tested for scale error at the test points specified in table I. During vibration, this unit shall be energized but not operated. After completion of this test, the unit shall be tested and the tolerance shall be specified for room temperature scale error."

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4.6.19 Line 2: Delete "Method 507" "MIL-STD-810"; add "Method 507.1" and "MIL-STD-810C."

4.6.21 Line 2: Delete "Method 508", "MIL-STD-810"; add "Method 508.1" and "MIL-STD-810C".

4.6.23 Line 2: Delete "Method 513.1," "MIL-STD-810"; add "Method 513.2" and "MIL-STD-810C".

4.6.24 Line 2: Delete "Method 516.1", "MIL-STD-810"; add "Method 516.2" and "MIL-STD-810C".

PAGE 21

4.6.26.3 Delete entire paragraph and add the following:

"4.6.26.3 Longevity test. Unless otherwise specified (see 6.2), two out of every six transducer tested for reliability shall be continued through a longevity test for 3,000 hours on each transducer. The same cycling test conditions shall apply. Testing time accrued on the two samples during reliability testing shall apply to the 3,000 hours. No accept/reject criteria for this test is specified. The evaluation shall be limited to the analysis of all failures, identification of design and pattern failures, and the submission of recommended design changes to the contracting activity.

4.7 Delete paragraph and add the new paragraph:

4.7. Inspection of packaging. The inspection of the preservation and interior package marking shall be in accordance with Group A and B quality conformance inspection requirements Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIl-STD-129.

Delete Section 5 in its entirety and add the following:

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, as specified (see 6.2).

5.1.1. Level A.

5.1.1.1. Cleaning. Transducer shall be cleaned in accordance with process C-1 or MIL-P-116.

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5.1.1.2. Drying. Trandsucer shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3. Preservation application. Not applicable.

5.1.1.4. Unit Packaging. Unless otherwise specified by the contracting activity, each transducer shall be packaged in quantity unit packs of one each in accordance with Method IIC of MIL-P-116. Each transducer shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the transducer when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2. Level C. Each transducer shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage durng shipment from supply to the first receiving activity.

5.1.3. Industrial. The Industrial preservation of transducer shall be in accordance with ASTM D3951.

5.2. Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1. Level A. Transducer packaged as specified in 5.1.1. shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2. Level B. Transducer packaged as specified in 5.1.1. shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting acitivty. Other requirements as specified in 5.2.1. apply.

5.2.3. Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4. Industrial. The packaged transducer shall be packed in accordance with ASTM D3951.

5.3. Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129."

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6.2 Step e : Delete "4.6.5.3" and add: "4.6.26.3".

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Custodian:
Air Force -99

Preparing activity:
Air Force -71

Project number:
6610-F144

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